

Unit 10, Lesson 1: Graphing Absolute Value Functions



Benchmark

MA.912.AR.4.3 Given a table, equation, or written description of an absolute value function, graph that function and determine its key features.



Learning Targets

- ☐ I can graph an absolute value function from a table or equation.
- ☐ I can determine key features of an absolute value function.



10.1.1 Warm-Up: Prototype Fabricator



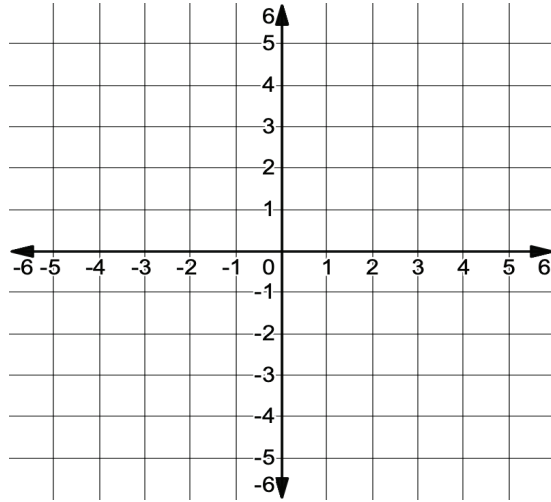
1. Jeff Tiedeken says, “A lot of times people come in the door and they don’t even have an idea of what they need. They just have a problem. So, the first thing is to understand the problem.”
 - a. Describe a time in any class where you first had to spend time to understand a problem.
 - b. Why do you think, for Jeff Tiedeken, understanding the problem is the first thing to do?



10.1.2 Guided Instruction: Graphing Absolute Value Functions

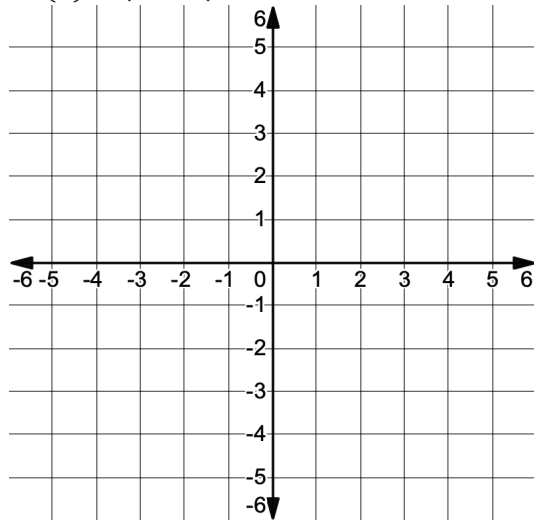
1. The table of values represents the absolute value function $f(x) = |x|$. Use the table of values to graph the function $f(x) = |x|$.

x	$f(x)$
-2	2
-1	1
0	0
1	1
2	2



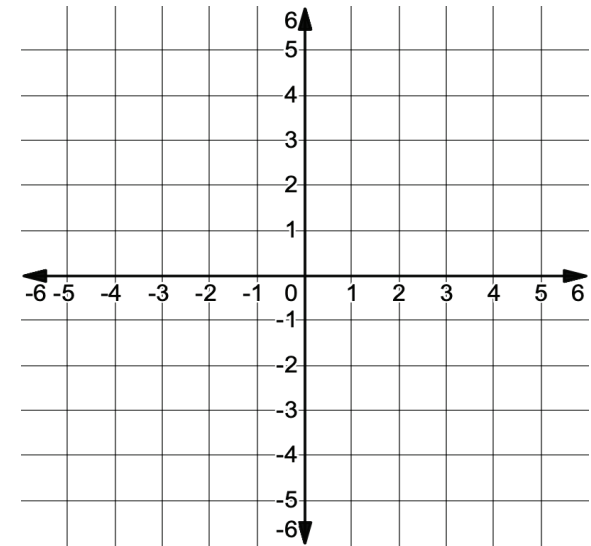
2. The table of values represents the absolute value function $h(x) = |x + 1| - 2$. Use the table of values to graph the function $h(x) = |x + 1| - 2$.

x	$h(x)$
-3	0
-2	-1
-1	-2
0	-1
1	0



3. Complete the table of values and graph the absolute value function $t(x) = |x - 1| - 1$.

x	$t(x)$
-3	
-2	
-1	
0	
1	
2	
3	



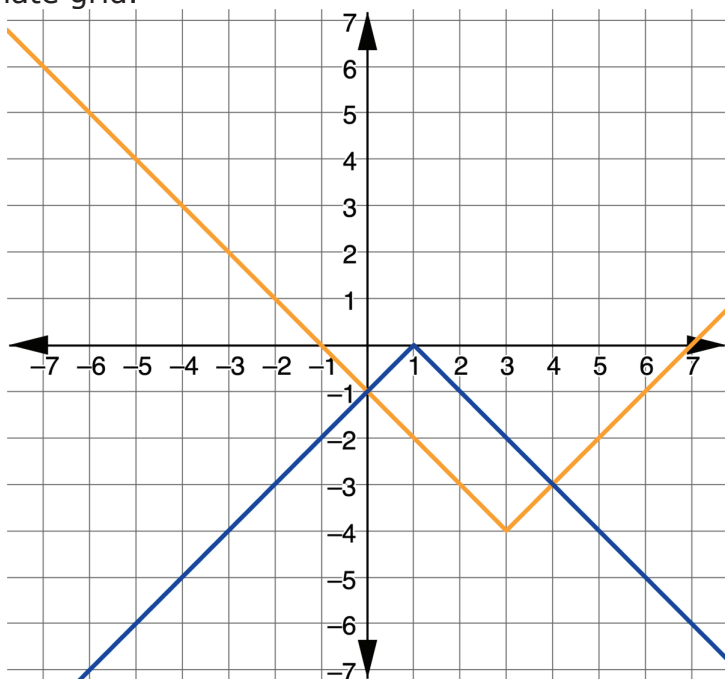
Absolute value is a number's distance from zero on a number line. Distance is expressed as a positive value.



10.1.3 Collaboration: Identifying Key Features of Absolute Value Functions

Similar to quadratic functions, absolute value functions have a vertex and a line of symmetry.

Two absolute value functions are shown on the coordinate grid.



Work with your partner to complete the following.

1. Complete the table for each function.

Function	Vertex	Equation of the Line of Symmetry
Blue		
Orange		

2. List two points on the orange graph that are equidistant from the line of symmetry.
3. List two points on the blue graph that are equidistant from the line of symmetry.
4. Work with your partner to identify additional key features of the specified function.

Function	x -intercept(s)	y -intercept	End Behavior
Blue			As $x \rightarrow \infty$, $y \rightarrow \underline{\hspace{1cm}}$. As $x \rightarrow -\infty$, $y \rightarrow \underline{\hspace{1cm}}$.
Orange			As $x \rightarrow \infty$, $y \rightarrow \underline{\hspace{1cm}}$. As $x \rightarrow -\infty$, $y \rightarrow \underline{\hspace{1cm}}$.

5. Complete the table for the orange function.

Key Feature	Inequality	Set-Builder Notation
Domain		
Range		
Increasing		
Decreasing		
Positive		
Negative		

6. Complete the table for the blue function.

Key Feature	Inequality	Set-Builder Notation
Domain		
Range		
Increasing		
Decreasing		
Positive		
Negative		

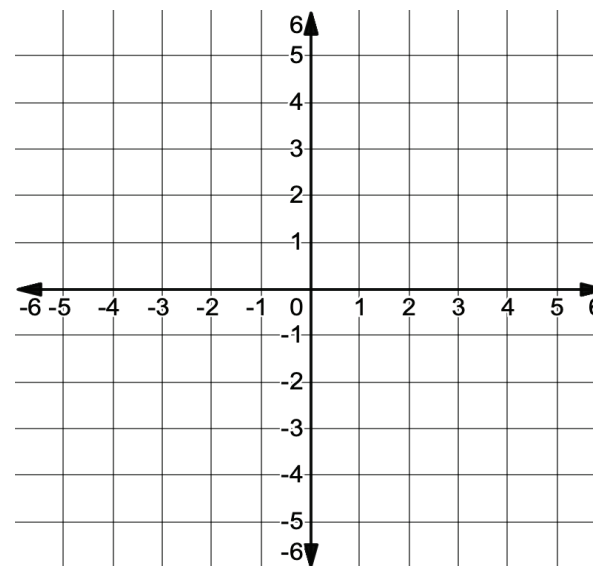


10.1.4 Guided Practice: Graphing and Identifying Key Features of Absolute Value Functions

Consider the function $g(x) = |x + 2| - 3$.

1. Complete the table of values to graph $g(x) = |x + 2| - 3$.

x	$g(x)$
-5	
-4	
-3	
-2	
-1	
0	
1	
2	



2. Label the vertex and draw the line of symmetry on the graph.

3. Complete the table.

x -intercept(s)	y -intercept
Right End Behavior	Left End Behavior

4. Use the inequality format to complete the table.

Domain	Range

5. Use set-builder notation to complete the table.

Interval(s) where the function is increasing	Interval(s) where the function is decreasing
Interval(s) where the function is positive	Interval(s) where the function is negative



10.1.5 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept.



I need help.



I got it.



I can help someone.

graphing absolute value functions			
x -intercepts			
y -intercept			
vertex			
line of symmetry			
domain			
range			
increasing/decreasing			
positive/negative			
end behavior			